

Geolocation Calibration for Rotational Interferometer based on Windowed Temporal Convolutional Multilayer Perceptron

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Abstract—Spaceborne passive geolocation has become increasingly critical in electromagnetic reconnaissance due to its wide-area coverage and strong concealment. However, geolocation performance often degrades under complex phase errors, particularly when systematic and random errors are coupled. To address this challenge, this paper proposes a geolocation error correction method for spaceborne rotational interferometers based on a Windowed Temporal Convolutional Multilayer Perceptron (WTC-MLP). The model takes raw phase difference observations and environmental parameters as inputs, applies a sliding-window strategy for sequence segmentation, and uses temporal convolution layers to capture local temporal context, enabling effective phase difference error calibration. Simulation results indicate that the proposed method improves geolocation accuracy by calibrating for systematic phase difference errors.

Keywords—spaceborne rotational interferometer, emitter geolocation, error calibration, windowed temporal convolutional multilayer perceptron

I. INTRODUCTION

Accurate and rapid emitter geolocation plays a critical role in modern electronic warfare and electromagnetic countermeasure scenarios, where reliable geolocation is essential for protecting communication and radar systems[1][2]. With the emergence of high-power, wideband, and intelligent jamming techniques, electromagnetic threats have become increasingly severe, making robust geolocation in complex environments a key research topic[3].

The spaceborne rotational interferometer has shown unique advantages for wide area emitter geolocation because its rotating baseline can provide multi angle observations[4]. However, its geolocation accuracy is affected by phase measurement errors, angular measurement errors, and random noise, which can lead to unstable or even ambiguous results[5]. Conventional geometric constraint and least squares iterative methods can reduce phase ambiguity to some extent[6], but their accuracy and robustness are still insufficient in complex scenarios where systematic and random errors are coupled.

Several calibration methods have been proposed, including grid based calibration tables[7], Gauss Newton

iterative estimation of systematic errors using multiple calibration sources[8], and self calibration schemes that jointly estimate emitter geolocation and phase bias from multiple ambiguous phase measurements[9]. However, these approaches often rely on prior conditions or involve relatively high computational complexity, limiting their applicability in dynamic and non-ideal environments. Recent advances in deep learning have enabled neural networks to capture nonlinear feature relationships in signal processing[10][11]. This provides an alternative pathway for calibrating phase difference errors.

Building on an analysis of the error mechanisms in spaceborne rotational interferometer geolocation, this paper proposes an error calibration method based on Windowed Temporal Convolutional Multilayer Perceptron (WTC-MLP). The proposed approach takes phase-difference observations with errors and environmental parameters as inputs. Since phase errors exhibit temporal correlation, weakly stationary behavior within a finite window, and dependence on environmental factors, a sliding window combined with temporal convolution is employed to capture local contextual dependencies in the time series, thereby enabling effective compensation of systematic errors. The method improves geolocation accuracy and solution stability, and it is meaningful for promoting the use of spaceborne interferometers in complex electromagnetic environments.

This paper is organized as follows. Section II presents the geolocation model. An error calibration method is proposed in Section III. Section IV showcases the outcomes of simulation results. Finally, Section V concludes this paper.

II. PROBLEM FORMULATION

As illustrated in Fig.1, the reference frame is the Earth-Centered Earth-Fixed (ECEF) coordinate system $O-XYZ$. In the interferometer rotation plane mounted on the satellite platform, the long baseline interferometer consists of two receiving channels forming the baseline $\overline{S_1S_2}$, which rotates about the baseline geometric center with angular velocity ω . During platform motion, the two channels intercept electromagnetic signals radiated by a stationary ground emitter T . The incident signal wavelength is denoted by λ , the baseline length be d , and the baseline-to-wavelength

ratio be $\kappa = 2\pi d/\lambda$, where the far-field condition $d \gg \lambda$ is assumed.

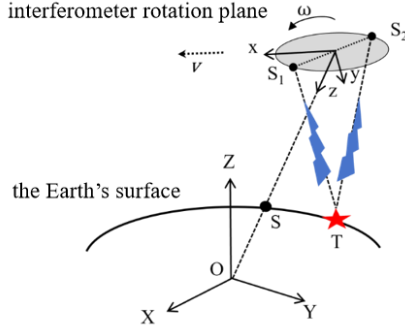


Fig. 1. Spaceborne rotational interferometer geolocation scenario.

At time instant t_i , the rotation angle of the rotational interferometer baseline is defined as $\theta_i = \omega t_i$. Then, in the satellite orbital coordinate frame $o-xyz$, the unit baseline vector is given by

$$\hat{\mathbf{d}}_i = [\cos(\theta_i), \sin(\theta_i), 0]^T \quad (1)$$

Given the satellite's high orbital altitude, the emitter can be regarded as a far-field source. Therefore, the interferometer rotation plane is approximated as being located at the satellite platform's center of mass. At time instant t_i , the position vector of the satellite platform's center of mass is denoted by $\mathbf{p}_i$, and that of the emitter T is denoted by $\mathbf{p}_T$. The unit position vector from the satellite observation platform to the emitter geolocation is denoted by

$$\mathbf{u}_i(\mathbf{p}_T) = \frac{\mathbf{p}_T - \mathbf{p}_i}{\|\mathbf{p}_T - \mathbf{p}_i\|} \quad (2)$$

Each pulse arrival instant during interferometer rotation is mapped to its corresponding rotation angle. By incorporating the platform's position and attitude information, the relative position vector and the associated coordinate transformation matrix can then be obtained. The measured phase difference φ_i and the wrapped phase difference φ_i^a of the rotational interferometer are then modeled as

$$\varphi_i = \varphi_i^{th}(\mathbf{p}_T) + \varepsilon_i = \kappa [\mathbf{u}_i(\mathbf{p}_T)]^T \mathbf{R}_i \hat{\mathbf{d}}_i + \varepsilon_i \quad (3)$$

$$\varphi_i^a = \text{mod}_{2\pi}(\varphi_i) \quad (4)$$

where φ_i^{th} denotes the theoretical phase difference, $\mathbf{R}_i$ is the rotation matrix from the satellite orbital frame to the ECEF frame at time t_i , and ε_i represents the phase-difference measurement error. The operator $(\cdot)^a$ indicates phase wrapping, and $\text{mod}_{2\pi}(\cdot)$ denotes the modulo- 2π operation.

Because phase is periodic, the measured interferometric phase difference is inherently modulo ambiguous, meaning that one wrapped observation can correspond to multiple possible unwrapped values. From a geometric perspective,

each candidate corresponds to a conical surface with the baseline as its axis and a half apex angle determined by the

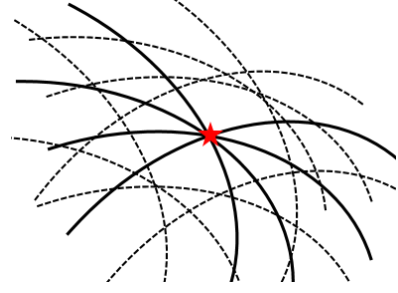


Fig. 2. The locus line cluster from phase difference measurements.

incidence angle. Intersecting the cone with the Earth's surface produces a geolocation locus line, and only one such line is consistent with the true emitter geolocation for a given observation. As the interferometer rotates and accumulates measurements, the true locus lines converge to a unique intersection point, while the ambiguous locus lines do not form a stable intersection, as shown in Fig. 2.

Assume that M phase-difference measurements are collected over a signal acquisition interval for a target within a designated region of interest. The measurement error is modeled as independent and identically distributed zero-mean Gaussian white noise. The unwrapped and wrapped phase-difference vectors are defined as

$$\boldsymbol{\varphi} = [\varphi_1, \varphi_2, \dots, \varphi_M]^T \quad (5)$$

$$\boldsymbol{\varphi}^a = [\varphi_1^a, \varphi_2^a, \dots, \varphi_M^a]^T \quad (6)$$

At time t_i , for an arbitrary candidate geolocation $\mathbf{p}$, cost functions with respect to $\boldsymbol{\varphi}$ and $\boldsymbol{\varphi}^a$ are defined as follows,

$$E(\mathbf{p}) = \frac{1}{M} \sum_{i=1}^M E_i(\mathbf{p}) = -\frac{1}{M} \sum_{i=1}^M \|\varphi_i - \varphi_i^{true}(\mathbf{p})\|^2 \quad (7)$$

$$E^a(\mathbf{p}) = \frac{1}{M} \sum_{i=1}^M E_i^a(\mathbf{p}) = -\frac{1}{M} \sum_{i=1}^M \|\text{mod}_{2\pi}(\varphi_i^a - \varphi_i^{true}(\mathbf{p}))\|^2 \quad (8)$$

For the cost function involving wrapped phase differences, a coarse-to-fine grid search strategy can be adopted[12]. The grid point that yields the maximum value of the cost function is then taken as the estimated target location.

III. ERROR CALIBRATION

A. Error Analysis

In the spaceborne rotational interferometer geolocation scheme, the dominant error sources can be broadly categorized as systematic errors and random errors which are summarized in Table I.

1) Systematic Errors

In the rotational interferometer geolocation scheme, systematic errors mainly consist of phase difference measurement errors and angular measurement errors. Amplitude and phase mismatches between the two receive channels on the interferometer baseline can introduce systematic errors in the measured phase difference. In addition,

environmental temperature variations may further exacerbate these errors.

TABLE I. ERROR CLASSIFICATION

Classification	Sources
Systematic errors	Phase measurement errors mainly arise from amplitude and phase inconsistencies among receive channels and environmental temperature variations.
	Angular measurement errors which are azimuth and elevation angle measurement errors cause attitude errors.
Random errors	Random phase difference errors caused by thermal noise in the received signal.

Angular measurement errors manifest as deviations in the azimuth and elevation angles of the observed line-of-sight direction. Denote the true incidence direction as (α, β) , where α represent the azimuth and β represent the elevation, respectively. In the presence of measurement errors, the observed direction is represented as (α', β') . The corresponding observation unit vector $\mathbf{b}'$ is

$$\mathbf{b}' = [\cos \alpha' \cos \beta', \sin \alpha' \cos \beta', \sin \beta']^T \quad (9)$$

Accordingly, the phase observation contaminated by angular errors can be expressed as

$$\varphi' = \mathbf{R}\mathbf{b}' \quad (10)$$

where $\mathbf{R}$ denotes the attitude transformation matrix.

2) Random Errors

In a rotating interferometer, the local oscillator signal exhibits phase noise. This noise is superimposed onto the received signal during the mixing stage, making it inherently stochastic. This error can be modeled as Gaussian-distributed noise, and the corresponding phase error satisfies

$$e \sim \mathcal{N}(0, \sigma^2) \quad (11)$$

where σ denotes the standard deviation of the phase noise.

B. Error Calibration based on Windowed Temporal Convolutional Multilayer Perceptron for Emitter Geolocation

To reduce the phase-difference errors analyzed in the previous section, a sliding window based temporal convolutional multilayer perceptron model for phase calibration is proposed. The overall calibration diagram is illustrated in Fig. 3.

The ideal phase difference of the k -th pulse is denoted by $\phi_k^* \in (-\pi, \pi]$, and the observed phase difference is denoted by $\phi_k \in (-\pi, \pi]$. The phase difference calibration term to be learned is defined as

$$\Delta\phi_k \triangleq \text{wrap}(\phi_k^* - \phi_k) \quad (12)$$

where $\text{wrap}(x) = \text{atan2}(\sin x, \cos x)$ constrains the phase difference to the principal interval $(-\pi, \pi]$. The network

outputs the predicted phase calibration $\Delta\phi_k$, and the corrected phase is then given by

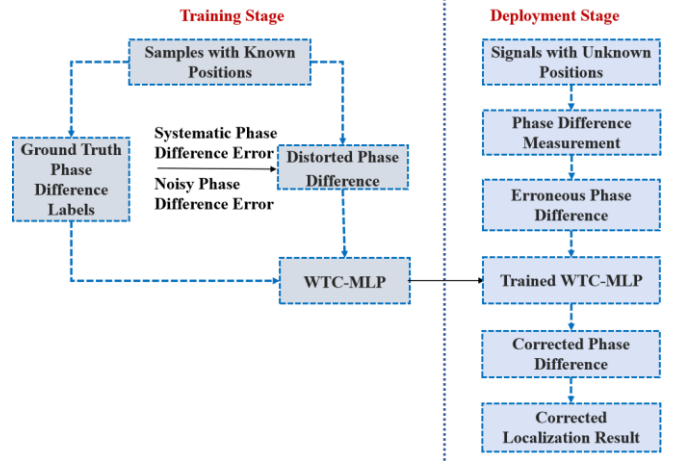


Fig. 3. Error calibration diagram.

$$\phi_k^{\text{corr}} = \text{wrap}(\phi_k + \Delta\phi_k) \quad (13)$$

By incorporating environmental data, a per-pulse feature vector is constructed. The network input is defined as

$$\mathbf{x}_k = [\sin \phi_k \cos \phi_k \sin \theta_k \cos \theta_k \text{lat}_k \text{lon}_k \text{alt}_k] \quad (14)$$

where θ_k denotes the antenna rotation angle, which is encoded by its sine and cosine components to avoid angular wrap-around ambiguity. The terms lat_k , lon_k , and alt_k are the standardized satellite latitude, longitude, and altitude, respectively. For downstream geolocation, the pulse sequence is segmented into fixed-length windows, where each window corresponds to one complete geolocation solution.

Phase errors often show correlation over time. In a finite window, their statistics change slowly, so they can be approximated as a weakly stationary process. The error also depends on the overall environment of the window, such as the satellite geolocation, attitude, and antenna rotation angle. This indicates that the error has contextual dependence. Therefore, within a sliding window of length W , explicitly modeling the local contextual relationship between neighboring pulses can improve the stability and accuracy of phase error calibration. Based on these properties, the paper proposes a window based on regression network with convolutional context awareness which is termed WTC-MLP. The network takes the feature sequence $\{\mathbf{x}_k\}_{k=t}^{t+W-1}$ within a window as input. It first uses multi-layer one-dimensional convolutions along the time axis to extract local correlation structures. A nonlinear regression head is then applied at each time index to output the phase calibration $\Delta\phi_k$. The network diagram is shown in Fig. 4.

The network input is $\mathbf{X} \in \mathbb{R}^{B \times T \times C}$ where B is the batch size, $T=W$ is the window length, and C is the feature dimension. The input is reshaped to $\mathbb{R}^{B \times C \times T}$ to match one-dimensional convolution. The convolutional feature extractor consists of three stacked one-dimensional convolutional neural network layers. The first layer expands the channel dimension from C to H . The next two layers keep the

channel dimension as H . The kernel size is $m = 3$, and same padding is used to keep the sequence length unchanged.

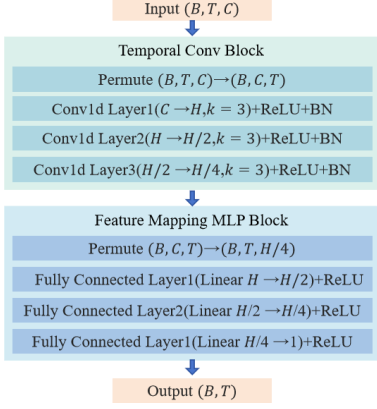


Fig. 4. WTC-MLP diagram.

Each convolution layer is followed by ReLU activation and batch normalization to strengthen nonlinearity and stabilize training. The output $\mathbf{F} \in \mathbb{R}^{B \times T \times H}$ is derived after convolutional encoding. The feature tensor $\mathbf{F}$ is subsequently fed into a per-time-step multilayer perceptron regression head, which shares parameters across time.

The multilayer perceptron reduces the feature dimension in three stages, from H to $H/2$, then to $H/4$, and finally to a scalar output. The final output is $\Delta\Phi \in \mathbb{R}^{B \times T}$. Since the convolution layers have a local receptive field, WTC-MLP aggregates information from neighboring pulses. Without introducing an explicit context branch, it still models local context within the window, which improves the continuity and robustness of phase error prediction.

IV. SIMULATION RESULTS AND ANALYSIS

A simulation scenario with four satellite orbits and three emitters is constructed in the local east–north–up (ENU) coordinate system, as shown in Fig. 5. The reference point is set at $(30^\circ, 115^\circ)$. The emitter locations in the ENU frame are given by $(649, -113)$ km and $(647, -161)$ km. Orbit 1 to orbit 3 observe emitters 1. Orbit 4 observe emitters 2. Each geolocation run uses 4000 consecutive pulses, corresponding to a full rotation cycle of the interferometer baseline. In the simulation, the azimuth error is set to 0.01° , the elevation error is set to 0.02° , and both the fixed phase bias and the standard deviation of the random phase noise are set to 2° .

The observations from orbits 1 to 3 corresponding to emitters 1 are designated as the training set, whereas the observations from orbit 4 corresponding to emitter 2 are designated as the test set. In the simulations, the network parameters are configured as follows. The batch size is set to $B = 16$ and the sliding window length is $T = W = 4000$. The input feature dimension is $C = 7$. The convolutional feature extractor consists of three one-dimensional convolutional layers with kernel size $m = 3$ and hidden channel dimension $H = 32$. The regression head reduces the feature dimension from H to $H/2$, then to $H/4$, and finally outputs a scalar at each time step. The circular Huber loss with a threshold of 0.1 rad is adopted for training.

The calibration results are shown in Fig. 6. The mean geolocation error before calibration is 5.8905 km, and it decreases to 1.0321 km after neural network calibration,

yielding an improvement of 82.48%. When the emitter is closer to the sub satellite point, the geolocation error is smaller. This agrees with the fact that the effective projection of the interferometric baseline increases and the angular sensitivity improves.

These results confirm that the proposed method can significantly reduce the impact of systematic errors and improve geolocation accuracy for rotational interferometers under cross orbit and cross target conditions.

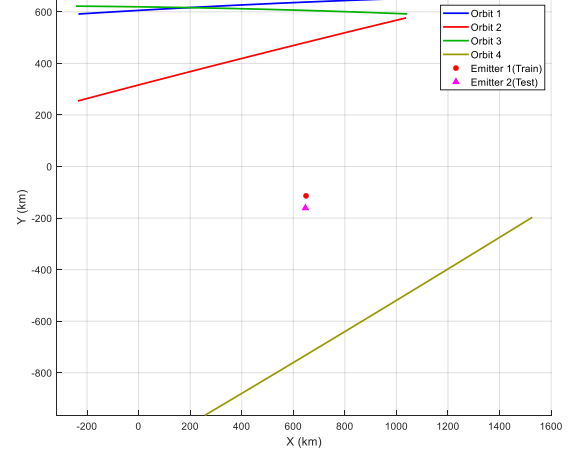


Fig. 5. Calibration scenario.

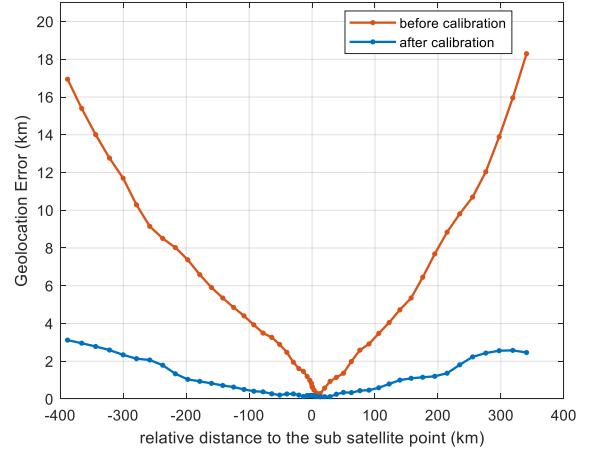


Fig. 6. Calibration result.

V. CONCLUSION

This paper addresses both systematic and random errors in spaceborne rotational interferometers for emitter geolocation, and proposes error calibration based on WTC-MLP. The proposed method jointly models phase observations and environmental parameters. By applying a sliding window with temporal convolution to capture local context in the time series, it provides effective calibration for phase difference errors. Simulation results show that the method significantly reduces geolocation error in multi orbit and cross target scenarios, achieving more than 80% accuracy improvement. It also shows good generalization and stability, and is suitable for engineering use in complex electromagnetic environments. Future work will focus on orbit real time calibration to improve adaptability and timeliness in practical spaceborne systems.

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